Problem Set 8

Due date: 11/7/2025

1. *(35 points)* Suppose that you are at the top of a rigid rocket which is half a light year tall. The rocket is accelerating such that your proper acceleration is $1g$.

   (a) *(2 points)* Show that $1g$ is very close to $1\text{lightyear}/\text{year}^2$, and use this value for g in this problem.

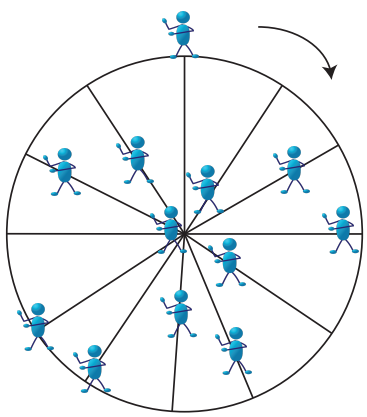
   (b) *(8 points)* What is the proper acceleration at the bottom of the rocket?

   (c) *(8 points)* Draw a spacetime diagram (in an inertial reference frame) that shows the front of the rocket, the back of the rocket, and at least three of the rocket's lines of simultaneity.

   (d) *(12 points)* For every second that passes for a clock at the bottom of the rocket, how much time passes for you?

   (e) *(5 points)* What is the proper acceleration of the bottom of the rocket if the rocket is 99/100 of a light year tall?

2. *(30 points)* Consider a set of rods of length R on a hub, spinning relative to some inertial frame:



   Builders are sent out onto the rods to build a platform on top of the rods while the whole structure is spinning. Assume each builder is stationed at just one place on the "wheel" and does all their work there. As shown in the diagram above, they are not all stationed at the ends of the rods. Take the outside of the wheel to be moving at $0.8c$ relative to the inertial frame.

   (a) *(10 points)* Suppose that the builders at a given distance D from the center measure the circumference of the circle they make on the platform. How does this circumference compare with the distance they measure to the hub ?

   (b) *(13 points)* Suppose that all of the builders were the same age in the morning before work. When they get together at the end of the day (as determined in the inertial frame), will they all still be the same age? If not, how will their ages differ as a function of their distance from the hub?

   (c) *(7 points)* We bring the spinning structure to a stop after the platform is built. Will it still fit together without having to buckle or tear? If not, what will happen to it?

3. *(60 points)* In this problem you'll derive the circumference of the relativistic merry-go-round using a different method, by finding coordinate system adapted for the reference frame of the edge of the relativistic merry-go-round.

   (a) *(3 points)* Starting with the inertial ground reference frame, write down the ground rest frame line element in spherical coordinates t, r, θ (you may work in three spacetime dimensions for this problem).

   (b) *(5 points)* If the merry-go-round is spinning clockwise at angular velocity ω, write down an equation for the worldline of an observer on the edge of the merry-go-round, parametrized by the time coordinate t in the inertial reference frame (i.e. write down an equation for $r(t)$, $\theta(t)$, etc).

   (c) *(12 points)* Define new coordinates $t' = t$, $r' = r(t)$, $\theta' = \theta(t)$. Write down the line element in terms of these primed coordinates. Use the line element to write down the components of the metric. Make sure that the metric that you write down is symmetric under the exchange of component indices.

   (d) *(10 points)* To compute the circumference, we must measure the length of the edge of the disk at a single moment of simultaneity. That means that we must integrate θ' from 0 to 2π at a moment of simultaneity. Explain why t' =constant is *not* a moment of simultaneity for the merry-go-round.

   (e) *15 points)* To find a plane of simultaneity for the disk, we need to synchronize all of the points around the disk. Let Alice and Bob be two observers located very close to one another on the edge of the merry-go-round. Alice's θ' coordinate is ${\theta'}_A$ and Bob's θ' coordinate is ${\theta'}_A + d\theta'$. We can synchronize Alice's and Bob's clocks by sending a light ray from Alice to Bob and then having Bob reflect it back to Alice. The observers can then synchronize their clocks using the halfway point of the trip. Let dt'_1 be the coordinate time elapsed for a light ray to travel from Alice to Bob, and let dt'_2 be the coordinate time elapsed for a light ray to travel from Bob to Alice. Alice computes half the trip time by taking the total time she waits for the light ray to arrive back and dividing it by 2. Bob computes the halfway point by measuring the time at which he receives the light ray. What are the coordinates of Alice's halfway point? What are the coordinates of Bob's halfway point? If these halfway points are to be identical, what must be the time lag dt' between Alice and Bob in terms of dt'_1 and dt'_2?

   (f) *(15 points)* Using the fact that $ds^2 = 0$ for a light ray, relate the expression in the previous part to $d\theta'$ in terms of ω and r. To compute the circumference, we now know that for every $d\theta'$ that we sum we must also add the dt' above. Use the expression you derived above to compute the circumference. Do you find $2\pi R\gamma$ like we did in lecture?